

THE DISTRIBUTION OF RESPONSIBILITY IN ENGINEERING

SAR Sambo, Institut de Technologie du Cambodge



OUTLINE

- The Problem of Many Hands
- Responsibility and the Law
- Responsibility in Organizations
- Responsibility Distributions and Technological Designs

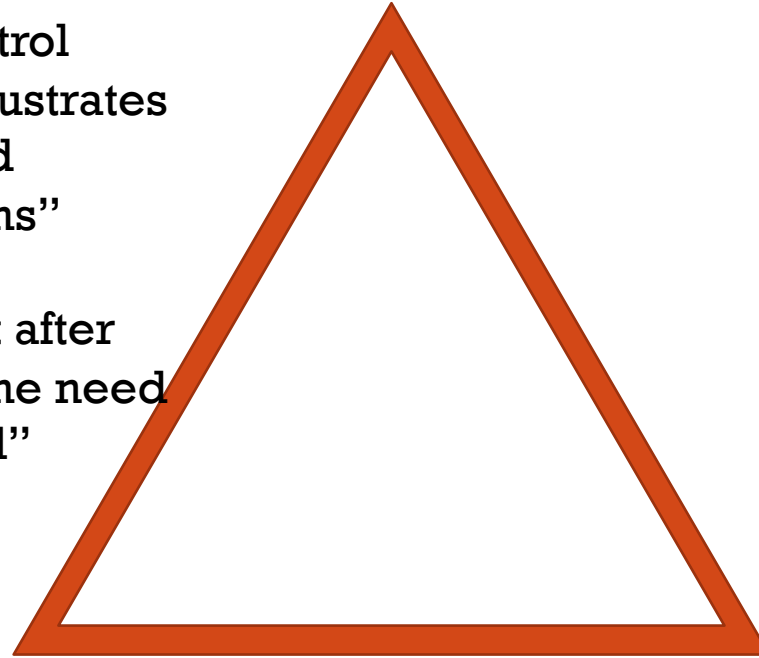
LEARNING OBJECTIVES

- Describe the problem of many hands and explain how it applies to engineering
- Examine responsibility distributions by the moral fairness and by the effectiveness requirement
- Explain the difference between moral responsibility and legal liability
- Distinguish different notion of legal liability and discuss their pros and cons
- Describe the different models for allocating responsibility in organizations to discuss their pros and cons, and to apply them
- Describe how engineering designs may affect the distribution of responsibility

THE PROBLEM OF MANY HANDS

Preventive dimension: a problem of prevention or as a problem of control

1. the problem of many hands “frustrates the need for compensation and retribution on the part of victims”
2. the “fact that no one can be meaningfully called to account after the event also means that no one need to feel responsible beforehand”



Practical dimension: It is often difficult in complex organizations or engineering projects to identify and prove who was responsible for what, especially for outsiders.

Moral dimension: it may turnout that nobody can reasonably be held morally responsible for an engineering disaster. This is morally problematic for at least two independent reasons:

1. If an engineering disaster occurs nobody can be held responsible.
=>**notion of collective responsibility.**
2. The second reason for attributing responsibility is the desire to learn from mistakes, and to do better in the future.

THE PROBLEM OF MANY HANDS



Problem of many hands is the occurrence of the situation in which the **collective** can reasonably be held **morally responsible** for an outcome, while **none of the individuals** can reasonably be held **responsible** for that outcome.



CAUSES OF THE PROBLEM OF MANY HANDS

Causes of the problem of many hands:

- The **distribution of information – foreseeable condition**
- The **conflict** between applying the responsibility conditions to the individuals and to the collective can also occur for **wrong-doing condition** condition
- Due to a combination of **conditions for responsibility**: the hierarchical nature of the organization and the specific procedures within the organization make an employee unable to act **freely**.

Dealing with the **problem of many hands** requires attention for the **distribution of responsibility** in engineering.

DISTRIBUTING RESPONSIBILITY

Distribute responsibility is the ascription or apportioning of responsibilities to various actors.

The resulting responsibility distribution can be evaluated based on:

- 1. Moral fairness requirement:** we consider it morally important to hold people responsible for their actions and the consequences of these actions if certain conditions are met.
- 2. Effectiveness requirement:** we want to avoid harm and stimulate desirable outcomes.

HOW SHOULD WE DISTRIBUTE RESPONSIBILITY?

Moral fairness requirement that a distribution of responsibility should be **fair (just)**.

- **Passive responsibility:** a person should only be held responsible if these four conditions are met:
 1. wrong-doing;
 2. causal contribution;
 3. foreseeability; and
 4. freedom of action.
- **Active responsibility:** a person should only be allocated responsibilities that **they can live by**.

HOW SHOULD WE DISTRIBUTE RESPONSIBILITY?

Effectiveness requirement is the moral requirement that states that responsibility should be so distributed that the distribution has the best consequences, that is, is effective in preventing harm (and in achieving positive consequences).

⇒ an ideal **distribution of responsibility** is both **morally fair** and **effective**.

HOW CAN WE RECONCILE THE REQUIREMENTS OF FAIRNESS AND EFFECTIVENESS?

Mechanisms for distributing responsibility and their moral fairness and effectiveness:

- law,
- organizational models for distributing responsibility and
- technological designs.

RESPONSIBILITY AND THE LAW

Legal responsibility = Liability

backward-looking responsibility according to the law. Usually related to the obligation to pay a fine or repair or repay damages.

DIFFERENCES BETWEEN LIABILITY AND MORAL RESPONSIBILITY

1. Conditions for liability are laid down in the law and may differ:

- for different types of actions,
- for different types of consequences and
- in different countries.

1. Conditions for moral responsibility

- Blameworthiness
 1. wrong-doing;
 2. causal contribution;
 3. foreseeability; and
 4. freedom of action.

It may be possible for a person to

- be morally responsible for an action while he or she is not liable.
- be liable without being morally responsible.

DIFFERENCES BETWEEN LIABILITY AND MORAL RESPONSIBILITY

2. Liability is established in an official and well regulated **procedure in court.**

- It requires a verdict by a judge or a jury and
- the liability conditions must be proven to apply in a formal juridical sense

3. Liability usually implies the obligation to pay a fine or to repay damages

4. Liability always applies after something undesirable has occurred.

2. Moral responsibility can be established more informally

3. Moral responsibility does not necessarily implies the obligation to pay a fine or to repay damages.

4. Moral responsibility is relevant both after the fact as well as before something undesirable has occurred.

DIFFERENCES BETWEEN LIABILITY AND MORAL RESPONSIBILITY

Table 9.1 Key differences between moral responsibility and legal liability

<i>Moral responsibility</i>	<i>Legal liability</i>
Moral blameworthiness based on conditions of wrong-doing, causality, freedom and foreseeability.	Based on conditions formulated in the law
Can be established more informally; you can also consider whether you are yourself responsible	Established in well-regulated procedure in court; juridical proof of conditions required
Not necessarily connected to punishment or compensation	Usually implies the obligation to pay a fine or to repay damages
Backward-looking and forward-looking	Backward-looking

DIFFERENCES BETWEEN LIABILITY AND MORAL RESPONSIBILITY

- Even if **liability** and **moral responsibility** are different notions, one might make an attempt to make them as similar as is feasible.
- One could for example base **liability** on the same conditions as **passive moral responsibility**.
- An argument in favor of basing liability upon moral responsibility is the consideration that it would be undesirable to have immoral laws.

=>If **morally irresponsible** behavior never leads to **punishment** on the grounds of **legal liability** then there remains little incentive to act morally so that few people will be encouraged to do that.

DIFFERENCES BETWEEN LIABILITY AND MORAL RESPONSIBILITY

- **Laws** that deviate in some respects from **morality** are not necessarily immoral.
 - One reason for this is that not everything that is legally allowed is also morally allowed
- ⇒ **Liability** could never be applied to all cases of **moral responsibility**.

LIABILITY VERSUS REGULATION

- **Regulation:** A legal tool that can forbid the development, production, or use of certain technological products, but more often it formulates a set of the boundary conditions for the design, production, and use of technologies.

LIABILITY VERSUS REGULATION

Legal liability

- deal with the social consequences of technology
- Liability may provide an attractive alternative **legal framework** for dealing with the social consequences of technology.
- Liability does not require the government to foresee the consequences of new technology
- but rather makes the ones developing those technologies, usually **companies** and the **engineers** employed by them, legally liable for those consequences under certain conditions.

Regulation:

- can forbid the development, production or use of certain technological products, but
- more often it formulates a set of the boundary conditions for the design, production, and use of technologies.
- regulations form a **regulative framework** for the design of a technology
- regulation will either have to forbid certain innovations or will lag behind the technological developments.

NEGLIGENCE

- **Negligence:** not living by certain duties.
- **Negligence** is often a main condition for legal liability. In order to show negligence for the law, usually proof must be given of
 1. a duty owed which is usually a **duty of care**,
 2. a breach of that duty,
 3. an injury or damage, and
 4. a causal connection between the breach and the injury or damage.

DUTY OF CARE

- **Duty of care:** the legal obligation to adhere to a reasonable standard of care when performing any acts that could foreseeably harm others.
- **Duties of care** typically arise in particular relationships such as between parent and child or between landlord and tenant.
- Also the relation between engineer and the public defines such a **duty of care**.
- For engineers, **the standard of care** both depends on what the public can reasonably expect from engineers and what is common practice in engineering.
- The **duty of care** is thus based on the sometimes implicit moral responsibilities of engineers and need not be made explicit in the law.

STRICT LIABILITY

- **Strict liability** does not require the defendant to be **negligent** in order to be liable.
 - It is usually enough that the defendant engaged in a risky activity and that this activity caused the damage done.
 - Technological innovation is obviously a risky activity in the sense that it might produce unknown hazards to society.
- ⇒ **Innovation** is a possible candidate for **strict liability**.
- ⇒ The **US** and the **EU** recognize **product liability**, which makes a manufacturer liable for defects in a product, without the need to prove that that manufacturer acted negligently.

STRICT LIABILITY

- **Strict liability** to technological products motivates engineers and the other people involved in innovation to be very careful by investigating possible hazards and taking precautions.
- ⇒ To meet **Mill's freedom principle** or the **principle of informed consent**.
- ⇒ Forbid subjecting people to (unknown) risks unless they have consented to the risks or the hazardous activity.

STRICT LIABILITY

Strict liability also has **disadvantages**:

1. **Strict liability** may well slow down the pace of innovation.
 2. It seems morally **unfair** to hold people liable when they are **not at fault** or **could not have foreseen** the damage.
 3. It also seems unfair to the potential victims that they have to bear the damage: they could have done even less than the engineers to prevent the damage.
- ⇒ The EU directive for **product liability** makes an **exception** for **developments risks**, defects that could not have been foreseen given **the state of scientific and technical knowledge** at the time the product was put into circulation.

BOEING CHARGED WITH 737 MAX FRAUD CONSPIRACY AND AGREES TO PAY OVER \$2.5 BILLION

- <https://www.justice.gov/opa/pr/boeing-charged-737-max-fraud-conspiracy-and-agrees-pay-over-25-billion>

CORPORATE LIABILITY

- **Corporate liability:** Liability of a company (corporation) when it is treated as a legal person.
- A main **advantage:** victims or the government do not need to find out which individuals in a company were responsible for a defect in a product but that they can simply sue the company as a legal person.
- If the company as a legal person is convicted it is bound to **pay a fine** or **compensate** the damage done.

CORPORATE LIABILITY

Advantages and limitations:

1. Corporations, unlike natural persons, do not possess a conscience.

⇒ legal instruments that are reasonably effective when applied to natural persons are not necessarily effective when it comes to corporations.

2. Most modern corporations are characterized by **limited liability**

=> The liability of shareholders for the corporation's debts and obligations is limited to the value of their shares.

3. Both the moral fairness and the effectiveness of corporate liability to a large extent depend on how the liability of the corporation is translated to individuals within the organization.

RESPONSIBILITY IN ORGANIZATIONS

Three different models for distributing responsibility in organizations:

1. **The hierarchical model** where those highest in the organization's hierarchy are held responsible;
2. **The collective model** in which each member of the organization is held to be jointly and severally responsible for the acts of the organization as a whole; and
3. **The individual model** in which each member of the organization is held responsible in relation to his or her contribution.

1. HIERARCHICAL RESPONSIBILITY MODEL

Only the **organization's top level** of personnel is held responsible for the actions of (people in) the organization.

The hierarchical model is attractive because of its **relative simplicity and clarity**.

1. HIERARCHICAL RESPONSIBILITY MODEL

1. In practice, the hierarchical responsibility **model is not always effective** in preventing undesirable consequences:
 - The managers of organizations may be, to an extent, outsiders within their own organization.
 - It is often very difficult for executives within an organization to get hold of the right information in time or to effectively steer the behavior of lower organizational units.
2. The knowledge that they will later have to account for damage done by the organization may motivate managers to gather the necessary knowledge and to better steer the organization.
3. The hierarchical responsibility model also seems somewhat **morally unfair**.

The allocation of responsibility along strict hierarchical lines may lead to moral objections.

2. COLLECTIVE RESPONSIBILITY MODEL

Every member of a collective body is held responsible for the actions of the other members of that same collective body (and for the responsibility of the collective).

- The **collective responsibility** model is not very attractive to **large organizations**,
- It is not possible to allocate responsibility in differing degrees to individual members of the collective.
- No one in particular tends to feel morally responsible for the consequences of the activities of the organization as a whole.
- The members of the collective must be able to effectively influence each other.

3. INDIVIDUAL RESPONSIBILITY MODEL

Each individual is held responsible insofar as he or she meets the conditions for individual responsibility.

- A main advantage in this model is that it is **morally fair**.
- The model also might seem **effective** because it encourages individuals to behave responsibly.
- **A disadvantage** of this model is that it may lead to **the problems of many hands**.
- One could try to avoid the **problem of many hands** through certain organizational measures, like:
 1. the better sharing of information or
 2. empowering individual employees so that they can live by their individual responsibility.

RESPONSIBILITY IN ORGANIZATIONS

- Which model can be best applied in a particular case partly depends on how the organization in question is actually organized.
- If an organization is, for example organized along strictly hierarchical lines, it will be both morally unfair and ineffective to apply the collective responsibility model.
- If a certain responsibility model is judged most desirable in a particular situation or for a particular task, attempts can be made to make the organization fit the responsibility model.

HOW SHOULD THE RESPONSIBILITY FOR SAFETY BE DISTRIBUTED?

An engineer for a company producing cars sets up a design team for the design of a new type of truck.

- how to **distribute the responsibility** for the **safety of the truck**.
- how to **allocate responsibility for safety** given a certain organizational set-up
- **formulating the desiderata** for an **allocation of responsibility** for safety:
 1. **All individuals** (or groups) to which a certain responsibility is allocated should be able to **live up to those responsibilities**.
 2. For **outsiders**, it should be clear whom to address if there is a concern or question about the safety of the truck.
 3. The **distribution of responsibilities** should be effective in the sense of **resulting in a safe truck**.

HOW SHOULD THE RESPONSIBILITY FOR SAFETY BE DISTRIBUTED?

Different models or combinations of models for allocating responsibility could be considered for achieving this:

1. The **hierarchical model**, in which the leader of the design team is responsible for safety;
2. The **collective model**, in which all design team members are equally responsible for the safety of the truck;
3. The **individual model** in which each member of the design team is responsible relative to his or her individual contribution;
4. The **responsibility for safety** can also be allocated to a special **safety official**. Many organizations, for example, have special HSE (Health, Safety, and Environmental) officers for this type of concern.

HOW SHOULD THE RESPONSIBILITY FOR SAFETY BE DISTRIBUTED?

The hierarchical model may do good on desideratum 2 (a clear address for outsiders), but it might be questionable whether the design team leader is able to oversee and steer all safety-related decisions, so that desiderata 1 and 3 are not fully met.

The individual model will motivate all design team members to take safety seriously. However, guaranteeing the safety of the entire truck may require an integral approach that takes into account the interaction between different parts of the truck that are designed by individual engineers (or sub teams). On the individual model, nobody may be responsible for this interaction, which may have disastrous results for safety, for which it is very difficult to hold any of the individual design team members responsible; hence the individual model may not meet desiderata 2 (address) and 3 (effectiveness).

On **the collective model**, the whole team (each individual member) can be addressed for failure to approach safety in an integral way. It is, however, questionable whether this model is effective (desideratum 3) and fair (desideratum 1) because the model presupposes that all design team members know and understand what the other members are doing and are able to influence that, which may be very difficult to attain in practice.

For such reasons, one could choose for appointing a **safety official** who is responsible for an integral approach (the fourth model), possibly in combination with the **individual model**, so that each team member is also responsible for his or her own contribution.

RESPONSIBILITY DISTRIBUTIONS AND TECHNOLOGICAL DESIGNS

- Engineering design influences the distribution of responsibility
- Design decisions also affect the passive responsibility for errors, for example in case of an accident.
- If the automatic pilot is designed in a way that it can only be turned on and off during take-off and landing,
- The human pilots do not longer have the freedom to correct the plane in case of a calamity during flight and can no longer be held responsible.
- If such a calamity results in a disaster because freedom to act is one of the conditions for responsibility
- ⇒ the designers (or producers) of the automatic pilot may rather be the ones that are to be held responsible.
- If the responsibility conditions can not be met for designers or producers
- ⇒ nobody can be held responsible, and a problem of many hands occurs.

RESPONSIBILITY DISTRIBUTIONS AND TECHNOLOGICAL DESIGNS

- If certain tasks are allocated to humans through design decisions, it should be ascertained that the conditions exist or can be created under which those individuals can responsibly carry out those tasks.
- Human pilots, for example, need information to be able to steer a plane.
- The system thus should provide them the right information on time.
- Note that even if this is the case, the pilots are dependent on the system, and on the system designers, for getting reliable information.
- Pilots cannot simply look out of the window of the plane and estimate the flight altitude.

RESPONSIBILITY DISTRIBUTIONS AND TECHNOLOGICAL DESIGNS

- **Technological design** may not only allocate responsibilities to individuals, but may also imply more complex divisions of labor and responsibility.
- **Some technologies** require a certain **social structure** to function properly.
- Philosopher of technology **Langdon Winner** has argued that the **atomic bomb** requires an authoritarian social structure:

"[T]he atom bomb is an inherently political artefact. As long as it exists at all, its lethal properties demand that it can be controlled by a centralized, rigidly hierarchical chain of command closed to all influences that might make its working unpredictable. The internal social system of the bomb must be authoritarian; there is no other way"

Technologies might require an allocating of tasks and responsibilities along hierarchical lines.